Unknown Whispers: Probing Cross-Lingual and Cross-Attack Generalization in Audio Deepfake Detection



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Chapter 1

1. Introduction

Deepfake audio generation refers to the synthesis of highly realistic speech using deep learning methods such as Generative Adversarial Networks, Variational Autoencoders, and Self-Supervised Learning models [13], [9], [4]. These technologies enable the creation of human-like voices from text inputs or by altering an existing speaker's voice, making them useful for applications in entertainment, accessibility, and personalized virtual assistants. However, the same technologies also introduce serious security risks, including voice phishing, misinformation, fraud, and unauthorized access to voice authentication systems.

As synthetic speech becomes increasingly indistinguishable from genuine human speech, there is a pressing need for detection methods that can reliably distinguish bona fide speech from manipulated audio. Current detection systems, such as RawNet2 [11], AASIST [2], and SSL-based models [12], have demonstrated strong performance on in-domain benchmarks. However, prior work has repeatedly shown that these systems degrade significantly when evaluated on out-of-domain data [5], [8]. This makes generalization a central challenge in audio deepfake detection.

A major difficulty in studying generalization is that multiple sources of domain shift often occur simultaneously. For example, a detector may be trained on one dataset and evaluated on another dataset that differs not only in synthesis algorithms but also in languages, speakers, recording conditions, and preprocessing pipelines. As a result, it becomes difficult to determine whether degradation is caused by novel attack types, linguistic variation, or other dataset-specific effects.

This work conducts a controlled analysis that separates two major axes of variation: synthesis-algorithm variability and language variability. In the first setting, we evaluate cross-lingual generalization by training on one language and testing on other languages while holding synthesis algorithms constant. In the second setting, we evaluate cross-generator generalization by training on one synthesis algorithm and testing on others while holding language constant. This design enables us to isolate the contribution of each factor to generalization failure.

Our findings indicate that the two factors do not affect detector performance equally. When synthesis algorithms are fixed, both RawNet2 and AASIST exhibit substantial cross-lingual transfer, although performance varies across training languages and architectures. In contrast, when language is fixed, and synthesis algorithms vary, the detector shows weaker and more unstable transfer. This suggests that synthesis-specific artifacts are a stronger source of domain shift than language identity in our experimental setup.

Motivated by this finding, we further evaluate whether domain generalization methods can improve robustness to unseen synthesis algorithms. We train models using Empirical Risk Minimization, Invariant Risk Minimization, CORAL, and GroupDRO across multiple source synthesis algorithms and evaluate them on held-out generators. This extension moves beyond diagnosis and tests whether multi-domain training objectives can mitigate the harder source of distribution shift.

The main contributions of this report are as follows. First, we present a controlled evaluation framework that disentangles language and synthesis-algorithm variation in audio deepfake detection. Second, we compare RawNet2 and AASIST under cross-lingual transfer while holding generators/synthesis algorithms constant. Third, we evaluate cross-generator robustness under controlled language conditions. Fourth, we introduce domain generalization experiments over held-out synthesis algorithms and analyze whether IRM, CORAL, and GroupDRO improve robustness relative to ERM.

Chapter 2

2. Related Work

2.1. Audio Deep Fake Detection

Audio deepfake detection aims to distinguish bona fide human speech from synthetic or manipulated speech. Early and contemporary anti-spoofing systems have relied on a range of acoustic, spectral, and neural representations. Recent end-to-end models have become increasingly common because they allow detectors to learn discriminative cues directly from waveform or time-frequency representations.

RawNet2 is an end-to-end anti-spoofing model that processes raw waveforms using convolutional and recurrent layers [11]. By avoiding handcrafted acoustic features, RawNet2 can learn temporal artifacts directly from speech signals. AASIST introduces an integrated spectro-temporal graph attention architecture to capture relationships across spectral and temporal regions, improving sensitivity to subtle spoofing artifacts [2]. SSL-based models such as wav2vec 2.0 have also been adapted for spoofing detection, often benefiting from strong pretrained speech representations [12].

Despite these advances, the central limitation remains generalization. Detectors trained on one distribution often perform poorly when exposed to unseen datasets, languages, speakers, recording environments, or generation methods. This has shifted the research focus from in-domain accuracy toward robustness under distributional shift.

2.2. Generalization in Audio Deep Fake Detection

Benchmarking generalization in deepfake detection has emerged as a central research direction. Existing studies can be broadly categorized into cross-dataset, cross-attack, and cross-language evaluations.

Large-scale benchmarks such as ASVspoof 2021 evaluate spoofing detection under more realistic and varied conditions [5]. Similarly, Müller et al. investigate whether generalization failures in audio deepfake detection are due to harder test samples or fundamentally different distributions [8]. These studies demonstrate that audio deepfake detectors can be unstable under

domain shift. However, cross-dataset evaluation often varies multiple factors simultaneously, making it difficult to isolate the cause of degradation.

Cross-attack studies examine whether detectors trained on one or more spoofing algorithms can detect unseen generation methods. Recent works, such as SpeechFake and SEA-Spoof, have highlighted that unseen generation methods remain a dominant source of degradation [1], [14]. However, in many evaluations, attack variation is coupled with changes in language, dataset source, or speaker distribution. Without controlling for these factors, it is difficult to determine whether poor performance is caused by the synthesis method itself or by confounding linguistic and acoustic variation.

Cross-language generalization has also received increasing attention. Marek et al. investigate whether audio deepfake detection models behave as "polyglots," but language changes are not always isolated from synthesis changes [6]. SpeechFake extends multilingual evaluation to many languages, but its use of multilingual pretrained representations introduces potential overlap between training and test languages [1]. This complicates the interpretation of cross-lingual transfer because strong performance may reflect adaptation within a multilingual pretraining space rather than genuine language generalization from scratch.

2.3. Separating Languages and Synthesis Variation

The present work is motivated by the need to disentangle linguistic and algorithmic sources of generalization failure. Prior work has shown that both language and attack type can affect detector performance, but these factors are often evaluated together. In contrast, our framework fixes one dimension while varying the other. When studying language transfer, all languages are restricted to a shared set of synthesis algorithms. When studying synthesis transfer, the language is fixed while synthesis algorithms vary.

This controlled design is important because it allows us to ask a more precise question: which source of variation is more damaging for detector robustness? If performance degrades primarily under language shift, then linguistic and phonetic coverage should become the main target for future improvement. If performance degrades primarily under synthesis shift, then the detector is likely relying on generator-specific artifacts and requires more synthesis-invariant representations.

2.4. Domain Generalization

Domain generalization aims to train models on multiple source domains such that they perform well on unseen target domains. Unlike domain adaptation, target-domain samples are not available during training. This makes domain generalization directly relevant to audio deepfake detection, where future spoofing algorithms may be unknown at deployment time.

Empirical Risk Minimization trains a model by minimizing the average loss over all training samples. It can overfit to spurious correlations present in the source domains. In audio deepfake detection, these spurious correlations may correspond to generator-specific artifacts rather than general properties of synthetic speech.

Invariant Risk Minimization attempts to learn representations for which the same classifier remains optimal across multiple environments. This encourages the model to preserve features that are stable across training domains while discarding domain-specific cues.

CORAL aligns feature covariance statistics across domains, encouraging representations from different source domains to occupy more similar distributions. GroupDRO optimizes for the worst-performing domain, reducing the model's tendency to perform well on easy domains while neglecting harder ones.

Chapter 3

3. Dataset and Experimental Setup

3.1. Datasets

To evaluate cross-lingual and cross-generator generalization, we use the Multilingual Audio Anti-Spoofing Dataset (MLAAD) [7]. MLAAD is a large-scale corpus of synthetic speech released under the Apache License 2.0 and designed to support multilingual audio anti-spoofing research. It contains approximately 486 hours of synthetic speech generated using 101 different TTS models across 40 languages and 42 architectures. These architectures include several widely used TTS families, including VITS, Tacotron2, and FastPitch [3], [10], [15]

MLAAD contains only synthesized audio. Therefore, bona fide speech is obtained from the M-AILABS Speech Dataset, which provides real speech for a subset of languages. Since M-AILABS provides real speech for English, Russian, German, Spanish, Italian, French, Polish, and Ukrainian, the cross-lingual experiments are constrained to languages for which both bona fide and synthetic samples can be paired. In our controlled cross-lingual setting, we use English, French, German, Italian, Polish, and Spanish, since these languages share a common set of synthesis algorithms in the filtered dataset.

The dataset construction follows two principles. First, each experiment is designed to isolate a single source of domain shift. Second, evaluation sets are constructed to avoid trivial overlap between training and test conditions. For cross-language experiments, synthesis algorithms are held constant while the language varies. For cross-generator experiments, language is held constant while synthesis algorithms vary. In the domain-generalization across-generators experiment, the language is kept constant.

3.2. Models

We evaluate RawNet2 and AASIST for the controlled cross-lingual experiments, while only using AASIST for the cross-generator experiment. RawNet2 processes raw audio waveforms using convolutional and recurrent layers, enabling end-to-end learning without handcrafted

acoustic features [11]. AASIST uses an integrated spectro-temporal graph attention mechanism to model time-frequency relationships and detect subtle synthetic artifacts [2].

For the domain generalization experiments, we use RawNet2 as the backbone. This allows us to compare ERM, IRM, CORAL, and GroupDRO under a consistent model architecture.

3.3. Evaluation Metrics

We report Equal Error Rate and accuracy. Equal Error Rate is the point at which the false acceptance rate equals the false rejection rate. Lower EER indicates better spoofing detection performance. Accuracy measures the proportion of correctly classified samples, with higher accuracy indicating better classification performance.

Although both metrics are reported, EER is treated as the primary metric because it is widely used in anti-spoofing and speaker verification evaluation. Accuracy can be sensitive to threshold choice and class balance, whereas EER provides a threshold-independent view of detector separability.

3.4. Training Protocol

For the original cross-lingual and cross-generator experiments, models are trained independently for each experimental condition. The final checkpoint is selected based on validation loss. Models are trained using the Adam optimizer and weighted cross-entropy loss with a batch size of 32. RawNet2 is trained with a weight decay of 0.0001. AASIST is trained with a base learning rate of 10^-4, a minimum learning rate of 5 x 10^-6, cosine annealing, and weight decay of 0.0001. Training is conducted using FP32 precision.

RawNet2 is trained under four objectives in the domain generalization experiments: ERM, IRM, CORAL, and GroupDRO. Each DG setting uses multiple source synthesis domains and evaluates on held-out synthesis domains that are not seen during training. All DG experiments are trained using the Adam optimizer with a learning rate of 10^-5, weight decay of 0.0001, and gradient clipping with a maximum norm of 5.0 for 10 epochs. IRM uses an invariance penalty weight of 100.0 with a 2-epoch annealing period. GroupDRO uses a group-weighting step size of 0.01. CORAL uses a correlation alignment penalty weight of 1.0 applied to the penultimate representation layer. All training is conducted using FP32 precision.

Chapter 4

4. Methodology

4.1. Cross-Language Generalization

To study linguistic transfer independently of attack coverage, we restrict all languages to the same set of synthesis algorithms. This ensures that performance differences across test languages are not caused by changes in the spoofing algorithms. We identify the maximal intersection of attack types shared by the selected languages, resulting in four common attack types: griffin_lim, multilingual bark, multilingual xtts v1.1, and multilingual xtts v2. These attack types are shared by English, French, German, Italian, Polish, and Spanish.

For each language, we train on that language and evaluate on all other languages. Each language-specific dataset contains approximately 8000 samples, with an approximately equal split between bona fide and spoofed speech. Since the synthesis algorithms are held constant, off-diagonal performance in the cross-language matrix measures the extent to which the detector transfers across languages rather than across attack types.

4.2. Cross-Generator Generalization

In this experiment, we fix the language and vary the synthesis generator. English is used for this setting because it provides high attack diversity in MLAAD. For each synthesis algorithm, we train a detector on samples generated by one particular algorithm and evaluate it on samples generated by other synthesis algorithms:

This produces a 16 × 10 cross-generator transfer matrix. Rows correspond to the synthesis algorithm used for training, and columns correspond to the synthesis algorithm used for testing. The matched train-test generator entries are excluded from the cross-generator transfer analysis where applicable. This setting tests whether detectors learn general spoofing cues or generator-specific artifacts. Strong off-diagonal performance would suggest robust, generator-invariant detection. Weak off-diagonal performance would indicate that the detector is overfitting to artifacts associated with the training generator.

4.3. Domain Generalization

In contrast to the cross-generator setting, where training is performed on a single synthesis algorithm, the domain generalization setting trains on multiple synthesis domains simultaneously and evaluates on held-out generators that are never observed during training.

To isolate synthesis variability independently of linguistic variation, the language is fixed to French across all domain-generalization experiments. We select six synthesis algorithms that provide sufficient paired bona fide and spoofed data, namely Facebook MMS TTS French, Griffin Lim, CSS10 VITS French, Tacotron2 DDC French, Multilingual Bark, and Multilingual XTTS v1.1. Each synthesis algorithm is treated as a separate domain.

We construct four DG splits, where each split trains on four source synthesis domains and evaluates on two held-out target domains. Each split contains approximately 8000 training samples and 4000 testing samples, with an approximately equal split between spoofed and bona fide speech. Each spoofed sample is paired with its corresponding original bona fide recording. Since the held-out synthesis algorithms are never observed during training, this setting directly evaluates whether the learned representation can generalize beyond the source generators rather than memorizing generator-specific artifacts. The DG splits we used are detailed in Table 1. The DG experiments are evaluated using per-domain accuracy and Equal Error Rate on the held-out synthesis algorithms.

Table 1: Domain generalization splits used in our experiments. Each split trains on four source synthesis domains and evaluates on two held-out synthesis domains.

Split	Training synthesis domains	Held-out test synthesis domains
Split 1	CSS10 VITS French; Tacotron2 DDC French; Multilingual Bark; Multilingual XTTS v1.1	Facebook MMS TTS French; Griffin Lim
Split 2	Facebook MMS TTS French; Griffin Lim; Multilingual Bark; Multilingual XTTS v1.1	CSS10 VITS French; Tacotron2 DDC French
Split 3	Facebook MMS TTS French; Griffin Lim; CSS10 VITS French; Tacotron2 DDC French	Multilingual Bark; Multilingual XTTS v1.1
Split 4	Facebook MMS TTS French; Griffin Lim; CSS10 VITS French; Multilingual XTTS v1.1	Tacotron2 DDC French; Multilingual Bark

Chapter 5

5. Results

5.1. Cross-Language Generalization with RawNet2

Figure 1 presents the RawNet2 cross-language EER matrix when the synthesis algorithms are held constant across languages. Rows correspond to the training language and columns correspond to the testing language, while diagonal entries are omitted because they represent the train-test language match.

The overall trend is a low off-diagonal EER across most language pairs with a mean value of 0.034, indicating that RawNet2 transfers reasonably well across languages under a fixed synthesis-algorithm setting.

The main outlier is German as a testing language. Across multiple training languages, testing on German results in noticeably higher EER compared to the other target languages. This indicates that German is the most difficult target language in the RawNet2 cross-lingual setting.

There are also a few isolated high-error cases outside German. In particular, Spanish-to-Polish and Italian-to-German produce some of the largest EER values in the matrix. However, these cases are exceptions rather than the dominant pattern. Overall, Figure 1 shows that RawNet2 maintains strong cross-language generalization in most settings, with German emerging as the clearest target-language outlier.

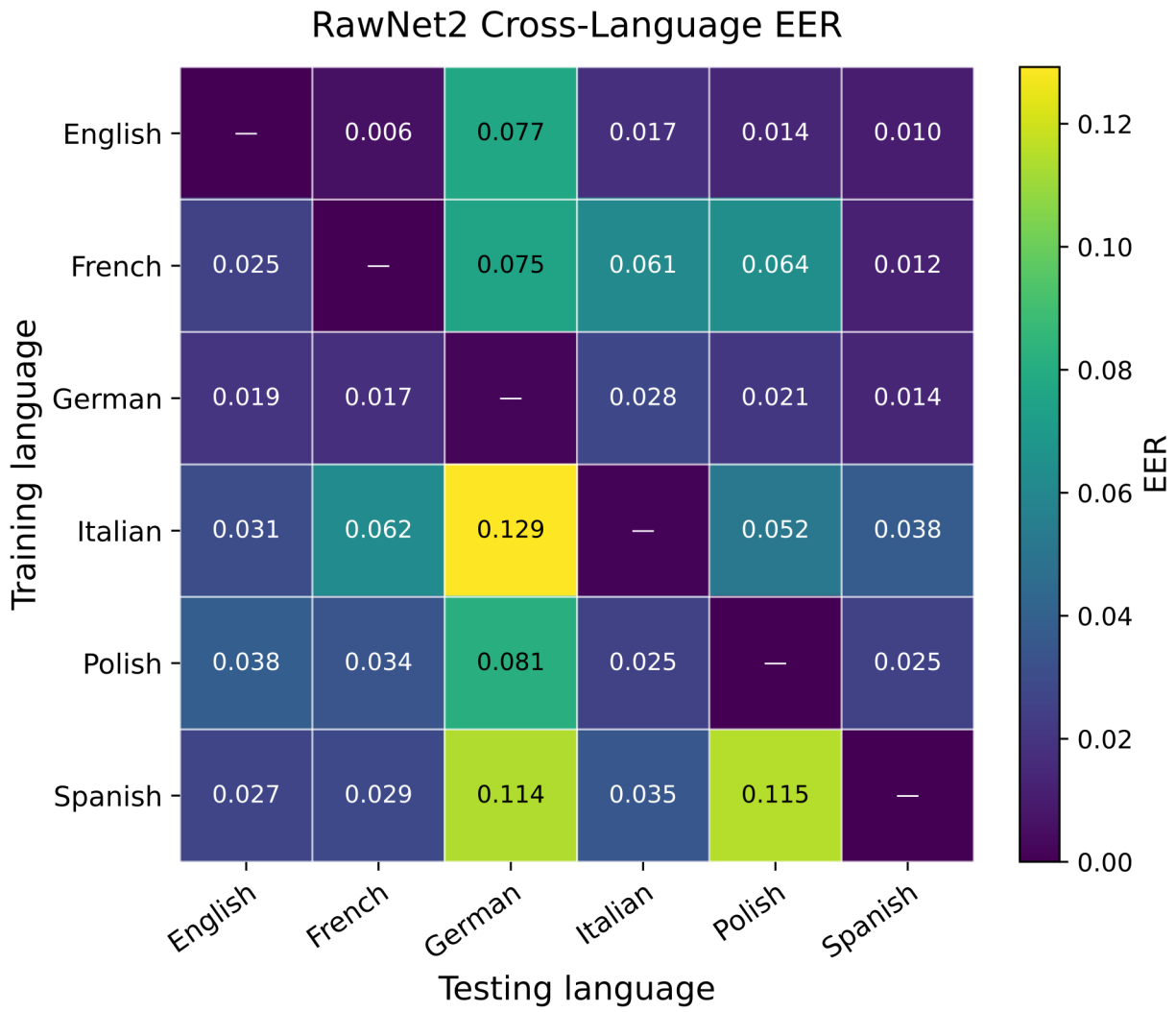


Figure 1: RawNet2 cross-language EER matrix under fixed synthesis-algorithm conditions. Rows represent training languages and columns represent testing languages. Diagonal entries are omitted because they represent train-test language matches.

5.2. Cross-Language Generalization with AASIST

Figure 2 presents the AASIST cross-language EER matrix under the same fixed synthesis-algorithm setting. As in Figure 1, rows correspond to the training language and columns correspond to the testing language. The overall trend is similar to RawNet2, with several off-diagonal entries remaining low, indicating that AASIST is also capable of cross-language

transfer when the synthesis algorithms are held constant. The mean EER value in this case is 0.035.

However, compared with RawNet2, AASIST shows a more unstable cross-lingual pattern. The EER values are higher across some language pairs, especially when the model is trained on Italian or Polish. German again appears as a difficult target language, with elevated EER values for English-to-German, French-to-German, Italian-to-German, Polish-to-German, and Spanish-to-German. This matches the RawNet2 trend, suggesting that German is consistently harder as a target language across both architectures.

The strongest outlier in AASIST is Spanish-to-Polish, with an EER of 0.214. This is substantially higher than the corresponding RawNet2 result, indicating a severe transfer failure for this language pair.

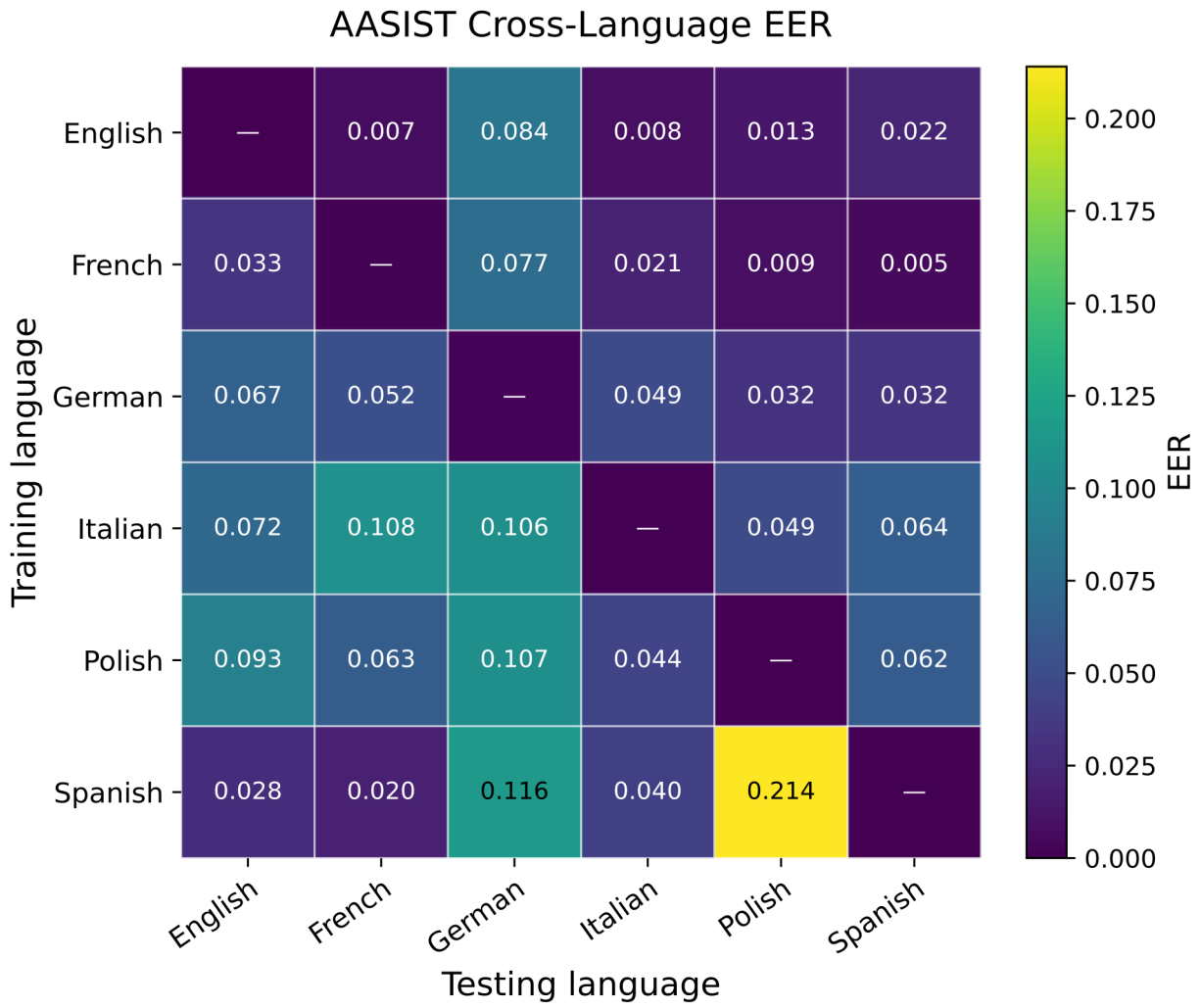


Figure 2: AASIST cross-language EER matrix under fixed synthesis-algorithm conditions. Rows represent training languages and columns represent testing languages. Diagonal entries are omitted because they represent train-test language matches.

5.3. Cross-Generator Generalization

Figure 3 shows the cross-generator EER transfer with AASIST. The mean EER for this cross-generator subset is 0.069, which is substantially higher than the mean cross-lingual EER observed for AASIST. This indicates that the synthesis-algorithm shift introduces a stronger generalization challenge than the language shift under our controlled setup.

The matrix still contains several low-EER transfer cases. For example, E2 TTS to F5 TTS, F5 TTS to E2 TTS, Mars5 to Griffin Lim, Mars5 to OpenVoiceV2, and OpenVoiceV2 to Mars5 all produce very low EER values. These cases suggest that some generators share detectable artifacts or acoustic patterns that allow the model to transfer successfully.

However, Certain testing generators produce consistently higher EER across many training generators. MeloTTS is the clearest example, with high EER values for several training generators, including E2 TTS, F5 TTS, Facebook MMS, MetaVoice, SpeechT5, and Parler variants. This suggests that MeloTTS behaves as a difficult target generator in this subset. Griffin Lim and Mars5 also produce high EER in some training conditions, especially when the model is trained on F5 TTS, Facebook MMS, MetaVoice, or OptiSpeech. These failures show that AASIST often does not learn a fully generator-invariant notion of spoofed speech.

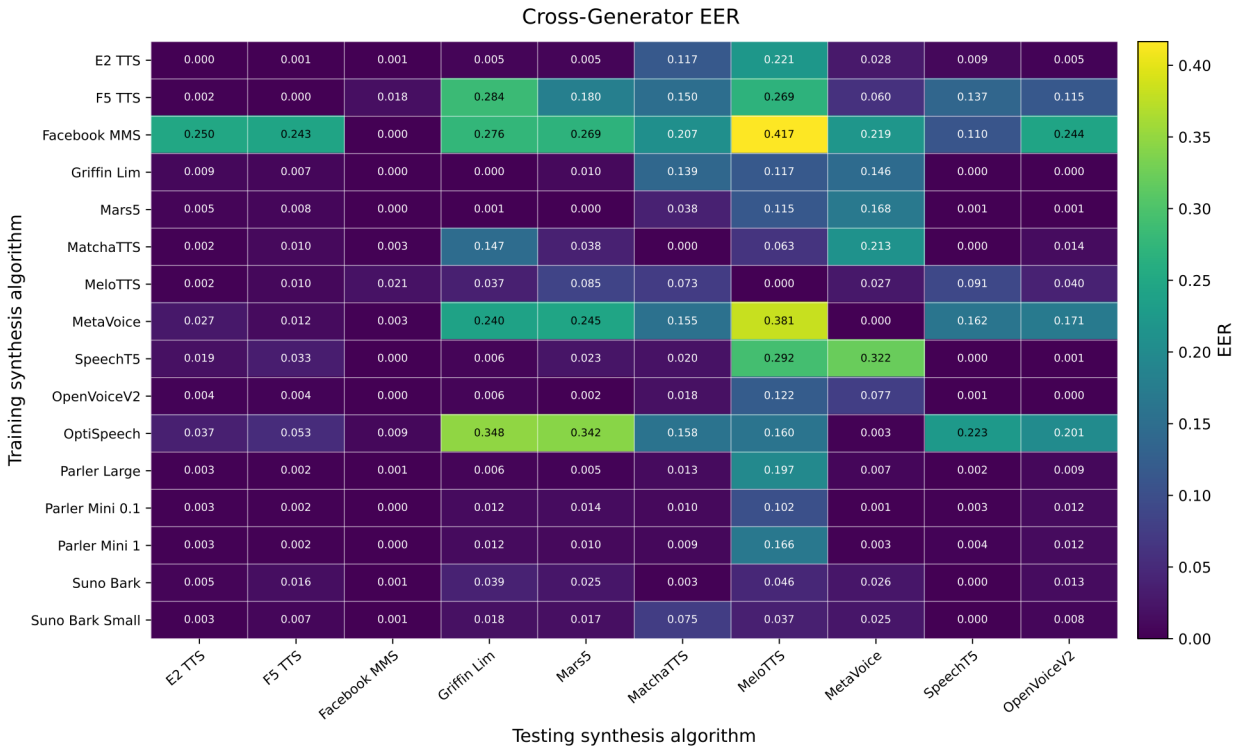


Figure 3: Cross-generator EER transfer matrix for AASIST. Rows represent the synthesis algorithm used for training, while columns represent the synthesis algorithm used for testing. Lower EER indicates better transfer to the target synthesis algorithm.

5.4. Domain Generalization Results

Table 2 shows the held-out synthesis-domain EER for ERM and the three domain generalization methods: IRM, CORAL, and GroupDRO. The overall results show that domain generalization provides partial improvements, but the gains are not consistent across all held-out generators. Table 3 shows that GroupDRO achieves the best average performance, with a mean EER of 0.3335 and a mean accuracy of 0.5899. CORAL also improves over ERM in terms of mean EER, reducing it from 0.3501 to 0.3374. In contrast, IRM does not improve average performance and produces a slightly higher mean EER than ERM.

Some held-out domains show clear improvement under domain generalization. For example, in Split 2, CSS10 VITS French has an ERM EER of 0.524, which drops substantially to 0.318 with CORAL and 0.306 with GroupDRO. Similarly, in Split 1, Facebook MMS TTS French improves from 0.114 under ERM to 0.107 under CORAL and 0.100 under GroupDRO. In Split 4, Multilingual Bark also improves slightly, with GroupDRO achieving the lowest EER of 0.254.

However, the results are not uniformly better than ERM. In Split 2, Tacotron2 DDC French performs best with IRM, reducing EER from 0.388 to 0.266, while CORAL and GroupDRO both perform worse than ERM on the same held-out domain. Similarly, in Split 3, GroupDRO improves performance on Multilingual Bark but performs worse than ERM on Multilingual XTTS v1.1. Griffin Lim also remains difficult for all methods, with EER values above 0.42 across ERM, IRM, CORAL, and GroupDRO. These results indicate that domain generalization helps in some held-out generator settings but does not fully resolve cross-generator degradation.

Table 2: Held-out synthesis-domain EER for ERM and domain generalization methods. Each split trains on four synthesis domains and evaluates on two unseen synthesis domains.

Split	Held-out synthesis domain	ERM	IRM	CORAL	GroupDRO
Split 1	Facebook MMS TTS French	0.114	0.156	0.107	**0.100**

Split 1	Griffin Lim	0.430	**0.425**	0.433	0.428
Split 2	CSS10 VITS French	0.524	0.596	0.318	**0.306**
Split 2	Tacotron2 DDC French	0.388	**0.266**	0.451	0.436
Split 3	Multilingual Bark	0.333	0.356	0.350	**0.318**
Split 3	Multilingual XTTS v1.1	0.367	0.363	**0.355**	0.407
Split 4	Tacotron2 DDC French	0.378	**0.368**	0.420	0.419
Split 4	Multilingual Bark	0.267	0.314	0.265	**0.254**

Table 3: Average held-out performance across all domain generalization test domains.

Method	Mean Accuracy	Mean EER
ERM	0.5873	0.3501
IRM	0.5858	0.3555
CORAL	0.5824	0.3374
GroupDRO	0.5899	0.3335

Chapter 6

6. Analysis

6.1. Phoneme Analysis

We compare the phoneme-overlap matrix with the cross-lingual EER matrix to examine whether linguistic similarity is related to cross-language transfer. A phoneme is the smallest contrastive sound unit in a language, and the phoneme-overlap matrix measures how much the phoneme inventory of one language intersects with another language. Since the diagonal entries correspond to train-test conditions using the same language, they are excluded from the correlation analysis. We only consider off-diagonal entries because these represent actual cross-language transfer. For each remaining language pair, we take the corresponding phoneme-overlap value and the corresponding EER value. This gives us paired values of the form: phoneme overlap versus cross-language detection error. We then compute Pearson and Spearman correlations between these paired values. Pearson correlation measures the strength of a linear relationship, while Spearman correlation measures whether the relationship is monotonic.

For EER, the Pearson correlation is -0.357 ($p=0.053$), while the Spearman correlation is -0.341 ($p=0.065$). The negative direction is expected as a higher phoneme overlap tends to correspond to lower EER. This suggests that language pairs with greater phonetic overlap may transfer better in the cross-lingual setting.

However, the EER correlations do not pass the conventional 0.05 significance threshold. Therefore, the result should be interpreted as a weak-to-moderate trend rather than strong evidence. It suggests that phoneme overlap may be associated with cross-language EER, but it does not fully explain the observed transfer behavior.

This is also clear from the German target-language results. German remains a difficult target language across multiple training languages, but this pattern is not fully explained by phoneme overlap alone. If phoneme overlap were the dominant factor, then German would only be difficult when paired with languages that have low phoneme overlap. Instead, its consistently

higher EER suggests that other generator-specific factors may be at play, such as prosody, acoustic realization, etc.

6.2. Generator Family Overlap Analysis

The current cross-generator set provides limited evidence for a generator-family effect. Some related generators, especially E2 TTS and F5 TTS, show strong mutual transfer. Both systems belong to the flow-matching-based, fully non-autoregressive TTS family. E2 TTS uses flow-matching-based mel-spectrogram generation, while F5 TTS extends this approach with a flow-matching Diffusion Transformer architecture [17], [16] .This suggests that shared synthesis characteristics may lead to shared detectable artifacts. However, our EER matrix does not include complete family blocks for Parler or Bark models, and many generators appear only as single representatives, appearing in the training set but not in the test set. Therefore, generator-family similarity may partially explain some low-EER transfer cases, but the current setup is not sufficient to establish family membership as the main determinant of cross-generator generalization.

6.3. Model Selection in Domain Generalization

While the experiments provide useful evidence about how ERM, IRM, CORAL, and GroupDRO behave under held-out synthesis-domain evaluation, they should not be treated as a definitive ranking of these algorithms. A central difficulty in domain generalization is that model selection is performed using source-domain validation data, while the final evaluation is conducted on unseen target domains. As a result, the checkpoint or hyperparameter setting that performs best on validation data may not necessarily generalize well to held-out synthesis algorithms.

Prior work on DomainBed argues that model selection is a core part of the domain generalization problem rather than a minor implementation detail. Different validation strategies make different assumptions about how source domains relate to unseen target domains, and inconsistent tuning practices can make algorithmic comparisons difficult to interpret [18]. In our experiments, we use fixed hyperparameters for ERM, IRM, CORAL, and GroupDRO rather than conducting a full hyperparameter search for each method. Therefore, poor performance for a particular method does not necessarily imply that the objective itself is ineffective; it may also reflect sensitivity to the chosen penalty weights, batch construction, learning rate, or checkpoint-selection strategy.

Chapter 7

7. Conclusions & Future Work

7.1. Conclusion

The results show that language shift is not the primary source of failure in our controlled setting. When synthesis algorithms are held constant, both RawNet2 and AASIST show cross-language generalization, with RawNet2 producing more stable transfer across language pairs. German appears as the clearest difficult target language, but the overall degradation remains limited compared to the synthesis-shift setting.

The cross-generator results show a much weaker generalization pattern. Even with the language fixed, performance varies substantially across synthesis algorithms, indicating that detectors often rely on generator-specific artifacts rather than fully synthesis-invariant spoofing cues.

Furthermore, the domain generalization experiments provide partial improvement but do not fully solve the problem. GroupDRO achieves the best average held-out EER, with CORAL close behind, while IRM is less stable across splits. However, the gains are inconsistent across held-out generators, and some domains remain difficult for all methods.

The main finding is that synthesis-algorithm variation is a stronger and more persistent source of distribution shift than language variation.

7.2. Future Work

Several directions remain open. Future work should first expand the language set beyond the European languages used in this study. A natural extension would be to study transfer from European languages to Asian languages. This would test whether detectors trained on languages such as English, French, German, Italian, Polish, and Spanish can generalize to languages with different phonetic inventories, prosodic patterns, and acoustic structures.

Moreover, future works could also evaluate a broader range of domain-generalization methods. The current results show that GroupDRO and CORAL improve average held-out EER, but the gains are inconsistent across generators. This suggests that standard DG methods are useful but noisy in this setting. Testing additional approaches, such as adversarial domain learning, meta-

learning-based DG, representation disentanglement, or domain-specific normalization, could provide a clearer picture of which objectives are better suited for audio deepfake detection.

Finally, studies should explore test-time adaptation in audio deepfake detection. Since unseen synthesis algorithms may introduce artifacts not present in the source domains, adapting the model at inference time could help reduce the gap between training and deployment conditions.

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